
UNIT 12 CONSTRUCTION OF COMPOSITE INDEX IN SOCIAL SCIENCES

Structure

- 12.0 Objectives
- 12.1 Introduction
- 12.2 Composite Index: The Concept
- 12.3 Steps in Constructing Composite Index
- 12.4 Dealing with Missing Values and Outliers
- 12.5 Methods to Construct Composite Index
 - 12.5.1 Simple Ranking Method
 - 12.5.2 Indices Method
 - 12.5.3 Mean Standardization Method
 - 12.5.4 Range Equalization Method
- 12.6 Principal Component Analysis (PCA)
 - 12.6.1 Conducting PCA and Analyses of Results
 - 12.6.2 Use of Output Indicators
 - 12.6.3 Use of Weight
 - 12.6.4 Limitations of Principal Component Analysis
- 12.7 Merits and Limitations of Composite Index
- 12.8 Let Us Sum Up
- 12.9 Exercises
- 12.10 Some Useful Books/References
- 12.11 Answer or Hints to Check Your Progress Exercises

12.0 OBJECTIVES

After going through this Unit, you will be able to:

- state the concept of composite index;
- describe the process of constructing the composite index;
- explain the various methods of composite index;
- discuss the merits and limitations of composite index; and
- learn how to interpret the results derived from composite indexes.

12.1 INTRODUCTION

In social sciences research, many a times the complex social and economic issues like child deprivation, food security, human well-being, human development etc. are difficult to measure in terms of single variable. The reason being that such issues have several dimensions and indicators. For example, it is difficult to explain the status of development of a district in terms

of a single variable because development is reflected in terms of several indicators. Some of such variables/indicators are quantitative type while others are of qualitative nature. In such situations, Composite index plays an important role to express the single value of several inter-dependent or independent variables. Further, the composite index makes it possible to compare the performance among region/states or districts etc. Hence, composite indexes are being recognized as a useful tool for policy analysis. Composite indexes can also be resorted to make comparison among different regions/sectors where wide range of variables are used.

In this Unit, we shall therefore discuss the concept of composite index, the process of their construction, various methods, their uses, and limitations. The study of the unit will also enable you to learn how to interpret the results. Let us begin to explain the concept of composite index.

12.2 COMPOSITE INDEX: THE CONCEPT

A composite index is an expression of a single score made by combination of different scores to measure a given variable or a group of variables. It expresses quantity or place (position) of multi facet aspects of a concept. The UNDP (2005) has explained that ‘a composite index expresses a quantity or a position on a scale of qualitative multi-faceted aspects..... which is relevant for information of the society’. An index can be a combination of independent indicators, or the average of an accumulation of selected indicators. The index represent specific concept or highlighting specific sector or areas like status of dalit (dalit deprivation index), food situation (food security/insecurity index), status of child (child deprivation/development index), quality of human development (human development index) etc.

The index that we construct is the outcome of some unidirectional variables or indicators. If an index is constructed by taking positive indicators, the higher value of index implies higher development and lower values imply lower development. For example, in case of index related to child development, if the variables are positive directional, the final index can be termed as ‘child development index’. On the other hand, if the variables are negative directional, it is called ‘Child Deprivation Index’. Let us take an example that child mortality is a negatively directional variable whereas percentage of children immunized is a positively directional variable. Such index is very useful in case of qualitative data. An index is more robust than a single indicator or variable.

The choice of indicator is a big challenge for researchers. The major issues in identifying measurable indicators are: whether data are available or not, whether we have reliable data, whether the data are cross-section or time series, and minimization of double counting raised from overlap or redundancy. Again if the variables to be chosen are easy to understand, they are more acceptable to a wider audience.

12.3 STEPS IN CONSTRUCTING COMPOSITE INDEX

The following steps are required to follow in construction of composite index:

Step	Why it is needed
1. Theoretical framework A theoretical framework need to be developed because it provides the basis for the selection and combination of variables into a meaningful composite indicator under a fitness-for-purpose principle (involvement of experts and stakeholders is envisaged at this step).	• To get clear understanding and definition of the multidimensional phenomenon to be measured. • To structure the various sub-groups of the phenomenon (if needed). • To compile a list of selection criteria for the underlying variables, e.g., input, output, process.
2. Data selection Indicators should be selected on the analytical soundness, measurability, country coverage, and relevance of the indicators to the phenomenon being measured and relationship to each other. The use of proxy variables should be considered when data are scarce (involvement of experts and stakeholders is envisaged at this step).	• To check the quality of the available indicators. • To discuss the strengths and weaknesses of each selected indicator. • To create a summary table on data characteristics, e.g., availability (across country, time), source, type (hard, soft or input, output, process).
3. Imputation of missing data Consideration should be given to different approaches for imputing missing values. Extreme values should be examined as they can become unintended benchmarks.	• To estimate missing values. • To provide a measure of the reliability of each imputed value, so as to assess the impact of the imputation on the composite indicator results. • To discuss the presence of outliers in the dataset.
4. Multivariate analysis Should be used to study the overall structure of the dataset, assess its suitability, and guide subsequent methodological choices (e.g., weighting, aggregation).	• To check the underlying structure of the data along the two main dimensions, namely individual indicators and countries (by means of suitable multivariate methods, e.g., principal components analysis, cluster analysis). • To identify groups of indicators or groups of countries that are statistically “similar” and provide an interpretation of the results. • To compare the statistically-determined structure of the data set to the theoretical framework and discuss possible differences.
5. Normalisation Should be carried out to render the variables comparable	• To select suitable normalization procedure(s) that respect both the theoretical framework and the data properties.

	• To discuss the presence of outliers in the dataset as they may become unintended benchmarks. • To make scale adjustments, if necessary. • To transform highly skewed indicators, if necessary.
6. Weighting and aggregation Should be done along the lines of the underlying theoretical framework.	• To select appropriate weighting and aggregation procedure(s) that respect both the theoretical framework and the data properties. • To discuss whether correlation issues among indicators should be accounted for. • To discuss whether compensability among indicators should be allowed.
7. Uncertainty and sensitivity analysis Should be undertaken to assess the robustness of the composite indicator in terms of e.g., the mechanism for including or excluding an indicator, the normalization scheme, the imputation of missing data, the choice of weights, the aggregation method.	• To consider a multi-modeling approach to build the composite indicator, and if available, alternative conceptual scenarios for the selection of the underlying indicators. • To identify all possible sources of uncertainty in the development of the composite indicator and accompany the composite scores and ranks with uncertainty bounds. • To conduct sensitivity analysis of the inference (assumptions) and determine what sources of uncertainty are more influential in the scores and/pr ranks.
8. Back to the data Is needed to reveal the main drivers for an overall good or bad performance. Transparency is primordial to good analysis and policymaking.	• To profile country performance at the indicator level so as to reveal what is driving the composite indicator results. • To check for correlation and causality (if possible). • To identify if the composite indicator results are overly dominated by few indicators and to explain the relative importance of the sub-components of the composite indicator.
9. Links to other indicators Should be made to correlate the composite indicator (or its	• To correlate the composite indicator with other relevant measures, taking into consideration the results of sensitivity analysis.

dimensions) with existing (simple or composite) indicators as well as to identify linkages through regressions.	To develop data-driven narratives based on the results.
10. Visualisation of the results Should receive proper attention, given that the visualization can influence (or help to enhance) interpretability.	- To identify a coherent set of presentational tools for the targeted audience. - To select the visualization technique which communicates the most information. - To present the composite indicator results in a clear and accurate manner.

Source: OECD (2008), 'Handbook on Constructing Composite Indicators Methodology and User Guide'

Caution on choosing variable

- 1) Whenever we choose any variable or a particular dimension for the index, we have to justify the inclusion of the variable into the index. This justification should come from empirical evidences or policy based research studies or from theoretical explanation.
- 2) If the variable is not unidirectional, the entire variables used should be converted to unidirectional. For example, in construction of the food security index, two variables like 'proportion of agricultural worker to total workers' and per capita value of agricultural output' is used for index. Here the first variable is a negatively directional whereas the second variable is positively directional. In this case we have to convert the entire variables into either positive direction or negative direction. If we want to convert this to positive direction, the first variable which has a negative direction should be deducted from 100. On the other hand, if we want to convert the entire variables into negative direction, we have to work out the reciprocal of per capita value of agricultural output.

12.4 DEALING WITH MISSING VALUES AND OUTLIERS

After selecting the indicators, you have to have a clear idea on missing values for each selected variable. Data can be missing in a random or non random fashion. In such situation, easy solution is to drop cases for which data is missing. However, before dropping the variable, you have to see the number of cases for which values are missing, because exclusion of such households for which data is missing could significantly lower sample size. Further, deleting such household may lead to some biases also. For example, in livelihood study of households, if there is a missing value, we have to find out the frequency of missing value because in many cases, the chances of missing value is high for lower economic class as compared to upper economic class. In such cases if we delete the households having missing value, it may result biases towards upper economic classes of household. In case, inclusion or exclusion of households having missing value put a little impact on the final result, we can delete such cases.

The second solution is to impute the missing value by applying some methodology or logic. But use of imputation is more common in academic indices and datasets. Here you can try to impute the missing value which should be as accurate as possible. For example, in calculating the consumption expenditure of 50 households of which about 5 cases are missing. In this case in imputing we can substitute the average value of consumption expenditure and replace the missing value by the averages. But for a more accurate estimation we can group these 50 households on the basis of value of assets holding and see to which asset category these 'MPCE missing' households belonged to. Then we can substitute the average consumption expenditure of the respected asset group.

Such approach has been used in a number of more recent indices such as the Corruptions Perceptions Index (CPI) developed by Transparency International and the World Bank's Worldwide Governance Indicators (WGI). Such approaches carry a dual advantage. They allow scores to be estimated for a maximal number of countries, and can use a broader range of indicators to triangulate indices for nebulous constructs.

Many a time, outliers may also disturb the analysis. For example if the income of four persons is 18000, 17000, 18500 and 19000 then the average is 18125. If we include the income of fifth person as 60000 then the average turns out to be 26500. In such matters, you have to drop the cases with high extreme value. Both in case of missing value and outlier you are expected to document and explain the selected imputation procedures and the results in detail.

Check Your Progress 1

- 1) What do you understand by the term Composite Index?

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- 2) Give some examples of Composite Index.

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- 3) List the steps involved in construction of Composite Index.

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- 4) State the alternative methods to deal with the missing values of the selected variables.

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12.5 DIFFERENT METHODS OF COMPOSITE INDEX

The various methods are used in construction of composite index. These include: Simple Ranking Method, Indices Method, Mean standardization method, Range Equalization Method, and Principal Component Analysis Method. Let us discuss one by one with simple examples.

12.5.1 Simple Ranking Method

Rank Method is one of the simplest methods to analyse the status of a region/district/state/country. In this method, the first step is to arrange/convert the variable into unidirectional for each district or state. Let us take an example of development status of different districts in Orissa state. First, we have to convert the relevant variables in either positive or negative direction. For example, the variable says 'the proportion of agricultural labour is negatively associated with the development of a region. Hence we have to convert this variable to positive direction by deducting this variable from 100 and change the labeling of variable as 'proportion of other than agricultural labour to total labour'. Higher the value of this variable, higher the development of the district or region. Depending on the value, each variable was ranked in the similar manner. Highest rank (1st) was given to the variables with high value and vice-a-versa. Alternatively, we can do reverse ranking of the variable i.e. highest rank is given to the variables with lowest value. The individual ranks are added to get the total rank value for the district. This has been illustrated in the example given in table 12.1. Our objective is to find out most backward tribal district in Odisha (having more than 50 per cent of tribal population to total population). A total of 11 districts qualify for tribal dominant area having 50 per cent or more proportion of population. We have selected 10 variables for identifying most backward tribal areas. Each of the variables for 11 tribal dominated districts was ranked according to total value of variable. The individual rank of all the variables of all the districts is given in table 12.2. As all the variable are unidirectional, rank 1 is given to district having higher value and vice-a-versa. After doing so we can add together the value of all the variables and find out the average rank for each district (Sum of total rank divided by number of variable i.e. 10). The average value of variable shows the status of district on development indicators.

Table 12.1: Districtwise Development Indicators of Tribal Concentrated Districts in Odisha

District	% of other than agricultural labour to all labour	% of net irrigated area to total sown area	Per capita value of agricultural output	Per capita consumption expenditure	Female Literacy rate	women work force participation rate	% of household getting safe drinking water	Percentage of villages having access to paved road	Percentage of villages having access to Phcs	Average casual wage rate
Actual Value										
Malkangiri	73	31	1304	201	18	67	82	21	16	36
Rayagada	50	23	479	201	18	67	78	34	13	37
Sundargarh	61	19	537	280	43	58	57	38	28	32
Nabarangapur	46	6	839	201	18	66	80	45	28	37
Kandhamal	62	14	410	201	33	68	32	21	19	36
Koraput	55	31	611	201	16	67	67	22	22	40
Mayurbhanj	60	30	572	280	35	62	44	42	21	31
Gajapati	52	25	529	331	24	77	43	29	20	34
Sambalpur	62	34	1075	280	50	62	56	31	13	39
Kendujhar	60	23	537	280	44	46	52	46	15	34
Jharsuguda	68	20	662	280	54	43	63	47	32	39
Average	59	23	687	249	32	62	60	34	21	36

Table 12.2: Districtwise Development Scenario of Tribal Orissa by Simple Ranking Method

District	% of other than agricultural labour to all labour	% of net irrigated area to total sown area	Per capita value of agricultural output	Per capita consumption expenditure	Female Literacy rate	women work force participation rate	% of household getting safe drinking water	Percentage of villages having access to paved road	Percentage of villages having access to primary health centre	Average casual wage rate	Average Rank/indices
Malkangiri	1	2	1	7	8	5	1	10	8	7	5.0
Rayagada	10	7	10	7	9	3	3	6	11	4	7.0
Sundargarh	5	9	7	2	4	9	6	5	3	10	6.0
Nabarangapur	11	11	3	7	10	6	2	3	2	5	6.0
Kandhamal	3	10	11	7	6	2	11	11	7	6	7.4
Koraput	8	3	5	7	11	4	4	9	4	1	5.6
Mayurbhanj	7	4	6	2	5	7	9	4	5	11	6.0
Gajapati	9	5	9	1	7	1	10	8	6	8	6.4
Sambalpur	4	1	2	2	2	8	7	7	10	3	4.6
Kendujhar	6	6	8	2	3	10	8	2	9	9	6.3
Jharsuguda	2	8	4	2	1	11	5	1	1	2	3.7

By this method, the most developed tribal district is Jharsuguda and most backward district is kandhamal.

12.5.2 Indices Method

The indices method is another simple method for calculating the status of development of an area/district or state. Like rank method, here also, at the first stage the variables are converted into one direction. After converting the variable into one (positive or negative) direction, we calculate the index. In this method we have to convert the district figures based on the average figure for the entire 11 districts as 100. Let us take an example of table 12.1. If the proportion of other than agricultural labour to total labour in malkangiri district is 73 and the average of that variable for all the 11 districts is 59, then the indices of that variable for Malkangiri district will be: $\frac{100}{59} \times 73 = 124$

In the same manner, we can find out the indices for all the variables for 11 districts as is given in Table 12.3. The final index of development for the districts can be obtained by taking the arithmetic mean for all the indicators (given in last column of table 12.3). After working out the average indices value for all the districts, we can rank all the districts. The district with highest indices will be most developed and the district with lowest indices will be least developed.

Table 12.3: Districtwise Development Scenario of Tribal Orissa by Indices Method

District	% of other than agricultural labour to all labour	% of net irrigated area to total sown area	Per capita value of agricultural output	Per capita consumption expenditure	Female Literacy rate	women work force participation rate	% of households getting safe drinking water	Percentage of villages having access to paved road	Percentage of villages having access to primary health centre	Average casual wage rate	Average Rank/indices
Malkangiri	124	133	190	81	57	108	138	62	78	99	107.0
Rayagada	85	99	70	81	57	108	131	98	64	103	89.6
Sundargarh	104	83	78	113	135	93	96	111	135	90	103.6
Nabarangapur	78	25	122	81	56	107	135	132	137	102	97.4
Kandhamal	106	59	60	81	102	109	54	60	92	100	82.3
Koraput	93	133	89	81	49	108	113	65	108	113	95.1
Mayurbhanj	101	129	83	113	109	101	74	123	103	86	102.2
Gajapati	89	106	77	133	76	123	73	86	95	95	95.3
Sambalpur	105	147	157	113	155	100	95	90	65	108	113.3
Kendujhar	102	99	78	113	136	74	88	136	71	94	99.1
Jharsuguda	114	88	96	113	167	69	105	138	153	109	115.3

In our example Kandhamal is the most backward with indices (82.3) and Jharsuguda (115.3) is the most developed district.

Let us make a comparison between the results obtained by rank method and indices method. Results in Table 12.4 shows that Jharsuguda, Sambalpur, and Malkangiri are most developed in both the methods whereas Raygada and Kandhamal are most backward in both the methods. While comparing both rank and indices method, we can also run correlation between the two set of indices. If the correlation is high then we can say that the finding in both methods is almost similar.

Table 12.4: A Comparison of Simple Ranking Method and Indices Method

District	Status of District in Rank Method	District	Status of District in indices Method
Jharsuguda	3.7	Jharsuguda	115
Sambalpur	4.6	Sambalpur	113
Malkangiri	5.0	Malkangiri	107
Koraput	5.6	Sundargarh	104
Sundargarh	6.0	Mayurbhanj	102
Nabarangapur	6.0	Kendujhar	99
Mayurbhanj	6.0	Nabarangapur	97
Kendujhar	6.3	Gajapati	95
Gajapati	6.4	Koraput	95
Rayagada	7.0	Rayagada	90
Kandhamal	7.4	Kandhamal	82

Note: Correlation of status of district in both rank and indices method is 0.927

12.5.3 Mean Standardization Method

Mean standardization Method is another simple method used both as a process of normalization and also a composite index. In this method we normalize the value of each variable and then work out the average of the normalized value for all the variables. The average of normalized value will be the composite index value. The normalization is done by dividing the actual value of variables by their respective means. Let us take an example from Table 12.1. For example the standardized value of the variable ‘% of other than agricultural labour to all workers’ for Malkangiri district can be found out by

Normalized value MS Method = Actual value (73)/Mean value (59) = 1.237

In the same process we can find out the normalized value of the entire variable for 11 selected districts. The standardized value of each selected indicator for different district is given in Table 12.5. The composite index value of each district given in the last column of Table 12.5 has been worked out by taking average of all the 10 selected indicators (row vector of the each district).